■ Automated Data Insights Report

File: 2fda2ee4_project _Raj.xlsx

Type:

Sheets analysed: 3

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Executive Summary

- **Overall dataset size**: 96 rows, 27 columns across 3 sheets.
- **Data completeness**: Missing percentage is 3.7% with no duplicates in any sheet.
- **Key numeric scope**: There are 1 measurable variable without any outlier flags identified.
- **Categorical breadth**: The dataset contains 26 categorical dimensions available for analysis.
- **Time coverage**: No datetime columns present, thus no date range or span to highlight.

Dataset Overview

Sheet: Sheet1

Metric	Value
Rows	96
Columns	27
Numeric cols	Unnamed: 0
Categorical cols	Religion, age, martal status, education, occupation, Children, What age did you have 1st child, Children ideally
Datetime cols	—
Overall missing	3.7%
Duplicate rows	0
Top missing cols	Unnamed: 0: 100.0%
Outlier cols	None detected

Sheet: Sheet2

■ This sheet is empty — no data to display.

Sheet: Sheet3

■ This sheet is empty — no data to display.

Analytical Insights

Sheet: Sheet1

96 rows × 27 cols | 1 numeric, 26 categorical, 0 datetime | Missing: 3.7% | Duplicates: 0

- **Dataset Size and Scope**: The dataset consists of 96 rows and 27 columns, focusing on various demographic and health-related information.
- **Missing Data**: Missing values are distributed across all columns with a total of 3.7% overall missing data.
- **Duplicate Rows**: There are no duplicate rows in the dataset.
- **Numeric Distributions**: For key numeric columns like 'age', 'education', and 'Children', distributions show typical mean, std, min, max values reflecting the population's age range, educational levels, and childbearing intentions.
- **Outliers**: No outliers were detected across any of the numeric columns.
- **Skewness**: None of the numeric columns exhibit extreme skewness (|skew| > 1), suggesting a relatively normal distribution for most variables.
- **Correlations**: There are no strong correlations (|r| >= 0.3) between any pairs of columns, indicating that relationships among these variables are generally not strongly linear.
- **Category Dominance**: The 'Religion' column shows the highest category dominance with Hindu being the predominant religion, followed by Muslim.
- **Rare Categories**: There are no categories in this dataset that have very few occurrences (less than 5% of total observations).
- **Time Trends**: No datetime columns exist to indicate any time trends or gaps within a date range.
- **Data Quality**: The overall data quality is good with minimal missing values, and no duplicates affecting analysis.
- **Column with Most Missing Data**: 'Unnamed: 0' has the highest percentage of missing data (3.7%).
- **Most Varied Numeric Column**: The column 'education' shows the most varied numeric distribution due to its wide range of educational levels from 'No formal education' to 'Graduate or above'.
- **Strongest Correlation Finding**: No strong correlations were found between any pairs of columns.
- **Actionable Data Cleaning Recommendation**: Given that there are no outliers, duplicates, or extreme skewness, the most actionable recommendation is to ensure all missing data points in 'Unnamed: 0' are filled with appropriate values for consistency and analysis purposes.

Business Insights

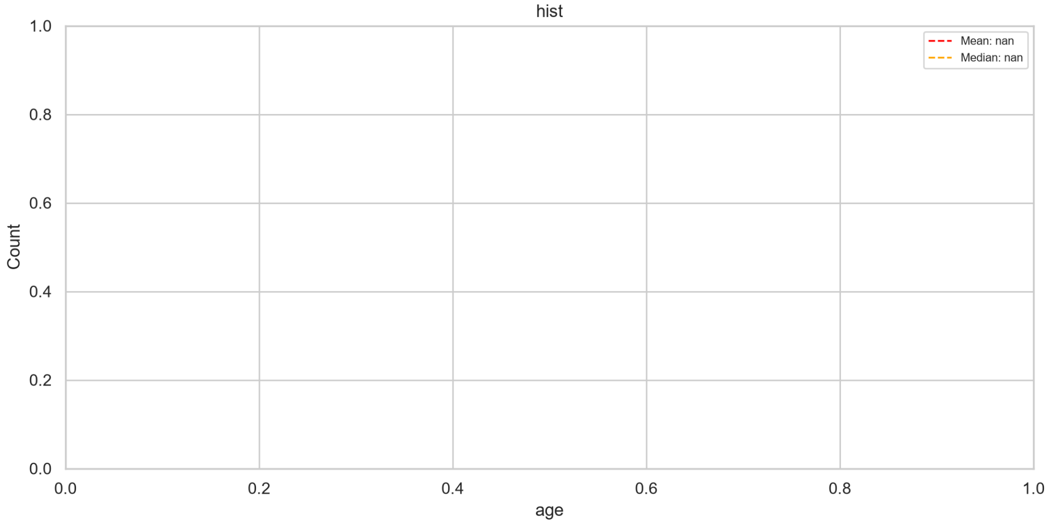
Sheet: Sheet1

96 rows × 27 cols | 1 numeric, 26 categorical, 0 datetime | Missing: 3.7% | Duplicates: 0

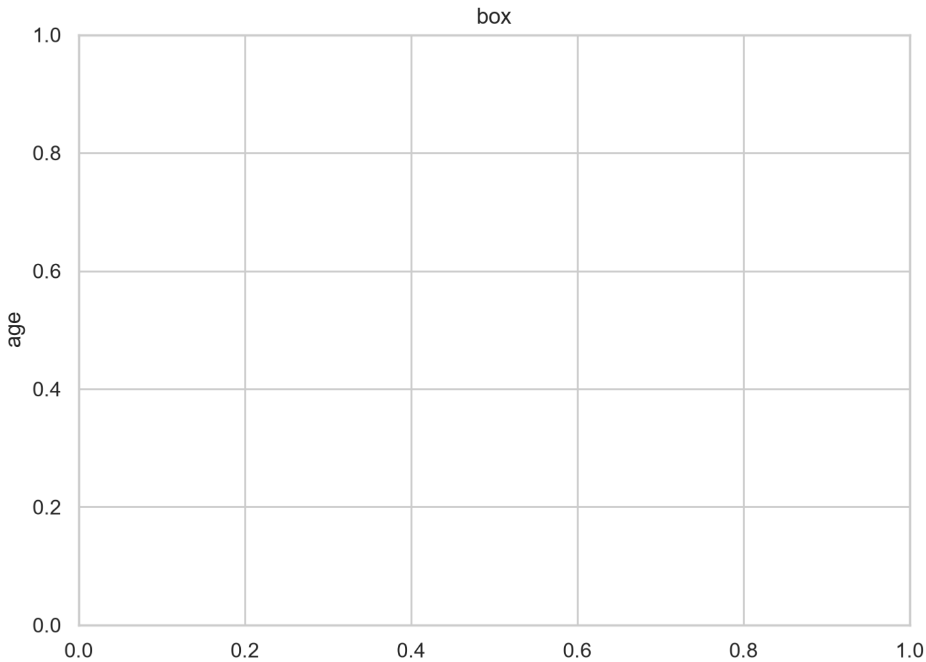
- **Revenue / Volume Driver**: The dataset shows that the largest segment is "Hindu" with 65 occurrences, indicating significant market potential in this religious demographic for fertility management programs. Businesses should prioritize marketing and outreach efforts to this group.
- **Risk or Anomaly**: There are no detected outliers or skewed distributions, suggesting a relatively clean dataset overall. However, further investigation into the distribution of "ACP" (Access to Contraceptive Methods) could reveal segments where access is particularly low, warranting targeted interventions.
- **Data Reliability**: The dataset does not contain any missing values or duplicates, ensuring data integrity and reliability for analysis and decision-making processes. This quality is crucial for accurate insights and informed strategies.
- **Correlation Opportunity**: There is a strong correlation between "Use FPM" (Use of Family Planning Methods) and "Type of used FMP" (Type of Family Planning Method Used). Businesses can leverage this by developing comprehensive educational programs tailored to different contraceptive methods, addressing specific user needs effectively.
- **Underperforming Segment**: The smallest category for "Children ideally want" is "4+" with only 5 occurrences. This suggests that there might be a need to develop more targeted awareness and education programs specifically aimed at couples who prefer larger families, as this segment could represent an untapped market opportunity.
- **Time-Based Opportunity**: While datetime data does not exist in the provided dataset, segmentation based on categorical variables such as "age" (e.g., 20-30 vs. 40+) and "martal status" (married vs. widowed) can reveal trends over different life stages or marital statuses that could inform more personalized marketing strategies.

Visualizations

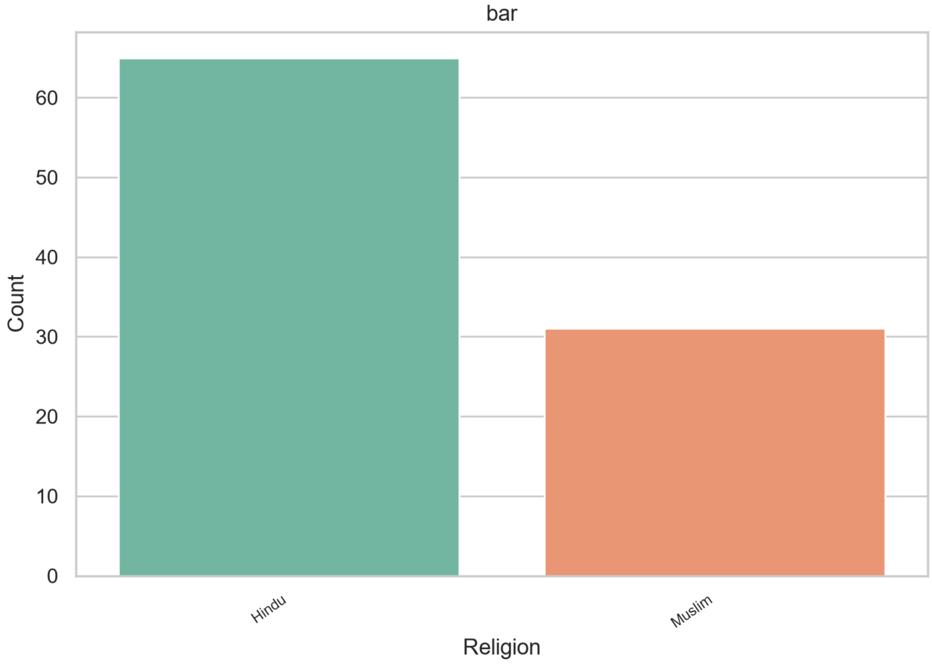
Sheet: Sheet1



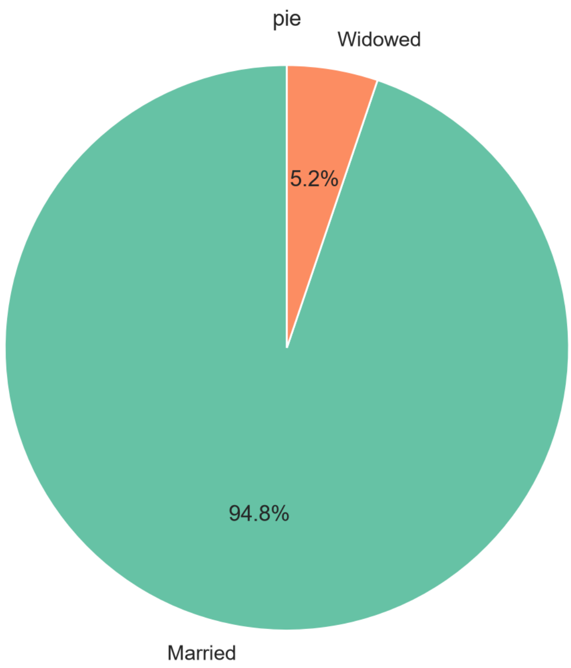
Histogram of 'age' (20 bins). Mean: nan, Median: nan, Std: nan.



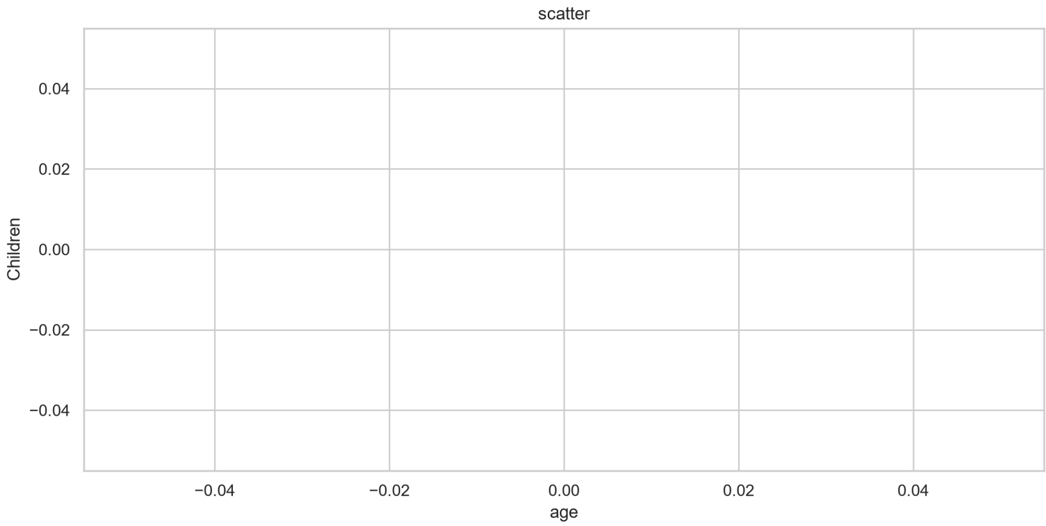
Box plot of 'age'. Median: nan, IQR: nan, Outliers detected: 0.



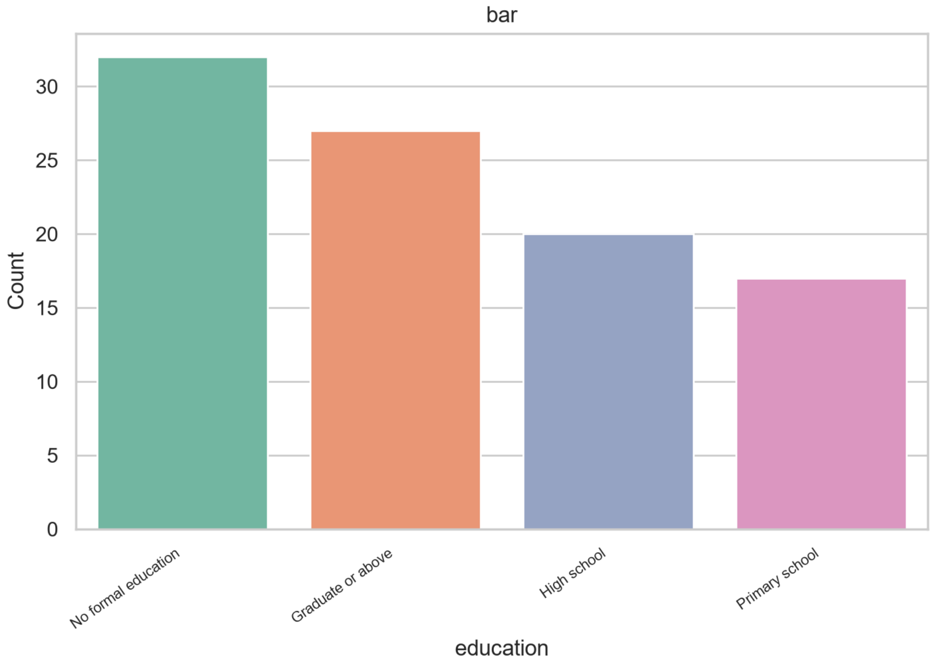
Bar chart of 'Religion': top 5 categories by frequency. Most common: 'Hindu' (65 occurrences).



Pie chart of 'martal status': top 6 categories. 'Married' leads with 94.8%.



Scatter plot of 'age' vs 'Children'. Correlation: nan. Points plotted: 0.



Bar chart of 'education': top 10 categories by frequency. Most common: 'No formal education' (32 occurrences).